An Introduction to Higher Algebraic K-Theory

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Chapter 1

Projective Modules

The subject of K-Theory concerns with studying vector bundles on spaces. Algebraic K-Theory studies the algebraic analogue of the same, projective modules over associative rings.

1.1 Projective Modules

1.1.1 Idempotence Criteria for Projective Modules

1.1.2 Constructing Projective Modules

- Patching along localizations
- Milnor Patching

1.1.3 Classification of Projective Modules

- Projective modules ¡-¿ vector bundles -

Chapter 2

The Grothendieck Group K_0

In this chapter, we will develop the notion of the Grothendieck Group K_0 accross a variety of contexts. First, we discuss the grothendieck group of a ring, then symmetric monoidal categories.

2.1 K_0 of a ring

Let R be a (unital) ring. The interesting objects of study will be finitely generated projective modules over R. We denote by $\mathbf{P}(R)$ the set of isomorphism classes of fintely generated projective modules over R. Note the $\mathbf{P}(R)$ derives a monoid structure through $\oplus$ because direct sums yield an associative operation that preserves isomorphisms. This monoid is also abelian due to the isomorphisms $M \oplus N \cong N \oplus M$ for any R-modules.

The idea is to consider the group naturally derived by implementing inverses for all elements in this monoid.

2.1.1 Group Completion of a Monoid

Given an (abelian) monoid $(M, +)$, we can derive an (abelian) group associated to M, which we shall call $\tilde{M}$, via the following procedure:

Define an equivalence relation on $M \times M$ by

$$(a, b) \sim (c, d) \Leftrightarrow a + d + e = b + c + e \text{ for some } e \text{ in } M.$$

This indeed gives us an equivalence relation. Let $\tilde{M}$ be the set of equivalence classes of $M \times M$ under this relation. This set admits a natural (abelian) group operation defined by

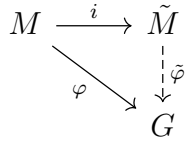
$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

This is a well-defined, abelian and associative operation on $\tilde{M}$ with the identity element $0 := [(0,0)]$ represented by any (l,l). Each element $[(a,b)]$ admits an inverse given by $[(b,a)]$ as $[(a,b)] + [(b,a)] = [(a+b, b+a)] = 0$.

A very familiar example of this construction is the abelian group $\mathbb{Z}$, which is the group completion of the abelian monoid $\mathbb{N}$ under $+$.

This construction is natural in M and follows a suitable universal property which we shall state.

Proposition 1 (Universal property of Group Completion)**.** *We have a monoid homomorphism $i : M \to \tilde{M}$ such that for any (abelian) group G, and a monoid homomorphism $\varphi : M \to G$, we have that φ must factor uniquely through i. That is, there is a unique map $\tilde{\varphi} : \tilde{M} \to G$ such that the following diagram commutes:*



Proof. The monoid homomorphism $i : M \to \tilde{M}$ is simply given by mapping a to $[(a,0)]$. This is indeed a homomorphism since $[(a,0)] + [(b,0)] = [(a+b,0)]$. Now suppose we are given $\varphi : M \to G$. Then $\tilde{\varphi} : \tilde{M} \to G$ is given by $[(a,b)] \mapsto \varphi(a) - \varphi(b)$, making the diagram commute. This map is unique since for any $a \in M$, we have $\tilde{\varphi}([(a,0)]) = \varphi(a)$, and consequently $\tilde{\varphi}([(0,a)]) = -\tilde{\varphi}([(a,0)]) = -\varphi(a)$, which altogether determines the map $\tilde{\varphi}$. □

Furthermore, any abelain group satisfying the given universal property is isomorphic to $\tilde{M}$ via a unique isomorphism.

There is an alternative (but equivalent) characterization of $\tilde{M}$ via the quotienting the free group on M by the monoidal relations.

Proposition 2. *Let $K(M)$ be the abelian group given by taking the free abelian group $F(M)$ on the set M and quotienting by all relations of the form $[a] + [b] - [a+b]$ where $a, b \in M$ and $[x]$ represents the element in $F(M)$ corresponding to the element $x \in M$. Then $K(M) \cong \tilde{M}$.*

Proof. We will show that $K(M)$ satisfies the universal property of $\tilde{M}$. Let $F(M)$ represent the free abelian group on the set M. The map $i : M \to K(M)$ on x is given by taking the equivalence class of the element x. Suppose $\varphi : M \to H$ is a monoid homomorphism to an abelian group H. Then define $\tilde{\varphi}([a])$ as $\varphi(a)$. Suppose $[a] = [b]$, then $[a] + [-b] = [0]$ so $a - b = 0$ and consequently $\varphi(a) - \varphi(b) = 0$. Thus the map $\tilde{\varphi}$ is well-defined

and clearly a group homomorphism. It is also unique since $\tilde{\varphi}$ is specified uniquely on the basis of $F(M)$. □

Example 2.1.1. *Let R be a (unital) ring satisfying the Invariant Basis Property (IBP). That is, $R^m \cong R^n$ implies $m = n$. Then the monoid of isomorphism classes of free R-modules with the operation $\oplus$ is isomorphic to $\mathbb{N}$, the isomorphism given by the map $R^n \mapsto n$. Consequently, its group completion is isomorphic to the abelian group $\mathbb{Z}$.*

Note that all fields (and division rings) F satisfy IBP since dimension of an F-module is independent of the choice of basis. Further, any ring R with a map $R \to F$, a field or a division ring F satisfies IBP. In particular, commutative rings all satisfy IBP.

Example 2.1.2. *Let X be a topological space. Then the set $[X, \mathbb{N}]$ of continuous maps $X \to \mathbb{N}$ forms an abelian monoid under pointwise addition of maps. Then the group completion of this monoid is isomorphic to the abelian group of continuous maps $X \to \mathbb{Z}$ denoted $[X, \mathbb{Z}]$. If X is quasi-compact, this is in fact a free abelian group. Notably, $[X, \mathbb{N}]$ also admits a commutative semiring structure via pointwise multiplication in addtion to the monoid operation. This makes $[X, \mathbb{Z}]$ into a commutative ring.*

2.1.2 Defining $K_0(R)$

Now we can define the group $K_0(R)$ as the group completion of the monoid $\mathbf{P}(R)$ consisting of finitely generated projective R-modules with the monoid operation given by $\oplus$.

Example 2.1.3. *Let F be a field or a division ring. Then all finitely generated projective F-modules are free. This gives us $K_0(F) \cong \mathbb{Z}$. More generally, for any local ring R, all finitely generated projective R-modules are free. This gives us $K_0(R) \cong \mathbb{Z}$.*

Example 2.1.4. *Note that finitely generated projectives over $\mathbb{Z}$ are free. Thus $K_0(\mathbb{Z}) \cong \mathbb{Z}$. More generally, for any ring R with the IBP*

2.1.3 Morita Equivalence and Invariance

Suppose we have (unital) rings R and S such that we have an equivalence of the respective categories of right modules:

$$\textbf{mod-}R \leftrightharpoons \textbf{mod-}S$$

Then we say that R and S are **Morita equivalent**.

A consequence of Morita Equivalence is that R and S yield isomorphic K_0 groups.

2.2 K_0 of a Symmetric Monoidal Category

2.3 K_0 of an Exact Category

Chapter 3

Groups K_1 and K_2 of a ring

Chapter 4

Constructions of Higher K-Theory

In this chapter, we shall explore the definitions of higher K-theory, for which we need to state some. The primary goal of higher K-theory is to define a 'K-Theory space' $K(R)$ for any ring R. We then define the higher K-groups to be $K_n(R) = \pi_n(K(R))$, and verify that this description matches with the classical theory for $n = 0, 1, 2$.

A number of approaches can be taken, all of which yield the same K-theory space, up to homotopy equivalence. In Section §1, we first recall some homotopy-theoretic notions. In Section §2, we describe Quillen's $+$-construction, introduced in []. In Section §4, we describe Quillen's Q-construction. Section §5 is devoted to showing the equivalence of the $+$ and Q-construction for the K-theory of finitely generated projective modules over a ring R.

4.1 Homotopy Theory Preliminaries

4.1.1 Geometric Realisation of a small category

Given a small category C, there is a way to interpret C as a CW-complex (and consequently as a topological space) such that the following properties are fulfiled:

We will note that the following homotopy-theoretic generalization of a topological group will be relevant when considering upcoming constructions.

Definition 1 (H-space)**.** *A H-space* [1] *X is a topological space with a binary operation $\mu : X \times X \to X$ and a distinguished element e such that*

[1] The H stands for Hopf

- $\mu(e, e) = e$
- $\mu(e, -) \simeq \mathrm{id}_X$
- $\mu(-, e) \simeq \mathrm{id}_X$

Additionally, we say that a H-space is homotopy-associative or homotopy-commutative if those respective properties hold up to homotopy. A detailed description is available in [].

Topological groups are trivially examples of H-spaces. A class of illuminating nontrivial examples come from n-th loop spaces $\Omega^n X$ of some based space X, where the multiplication is given by concatenation of paths.

The (path) connected components of a homotopy-associative H-space determine a group.

We now describe the notion of a homtopy fiber, which is useful for organizing how homotopy groups behave across a map $f : E \to B$.

Definition 2 (Homotopy Fiber)**.** *Let $f : E \to B$ be a map of based spaces. The homotopy fiber $F(f)$ of this map is the homotopy-pullback of the cospan $E \to B \leftarrow PB$, where PB is the (based) space of paths in Y.*

More concretely, we can define $F(f)$ to be consisting of pairs (e, γ) where $e \in E$ and γ is a path in B starting at the basepoint such that $f(e) = \gamma(1)$. We may declare $F(f)$ to have the basepoint $(*_E, *_{PB})$,

The key property of the homotopy fiber is that the sequence $F(f) \to E \to B$ of based space maps which yields a long exact sequence:

$$\begin{aligned}\cdots \to \Omega^{n+1}B \to \Omega^n F(f) \to \Omega^n E \to \Omega^n B \to \cdots \\ \cdots \to \Omega F(f) \to \Omega E \to \Omega B \to F(f) \to E \to B\end{aligned}$$

which in turn yields a long exact sequence of groups/pointed sets:

$$\begin{aligned}\cdots \to \pi_{n+1}B \to \pi_n F(f) \to \pi_n E \to \pi_n B \to \cdots \\ \cdots \to \pi_1 F(f) \to \pi_1 E \to \pi_1 B \to \pi_0 F(f) \to \pi_0 E \to \pi_0 B\end{aligned}$$

Additionally, if E and B are H-spaces, then the last few terms respcet the induced group multiplication.

4.1.2 +-construction on a topological space

Definition 3. *A space X is said to be acyclic if it has the homology of a point, i.e., $H_*(X, \mathbb{Z}) \cong H_*(*, \mathbb{Z}) \cong \mathbb{Z}$.*

Definition 4 (Acyclic maps)**.** *Let $f : X \to Y$ be a map of based connected CW-complexes. We say f is acyclic if the homotopy fiber $F(f)$ is acyclic, i.e., has the homology of a point.*

Note that an acylic map induces an isomorphism on integral homology $f_* : H_*(X, \mathbb{Z}) \cong H_*(Y, \mathbb{Z})$. Furthermore, we have the following result characterising acyclic maps:

Lemma 1. *Let X, Y be connected CW-complexes. A map $f : X \to Y$ is acyclic if and only if for every $\pi_1(Y)$-module M, it induces an isomorphism $H_*(X, M) \cong H_*(Y, M)$.*

Definition 5 (relative +-construction)**.** *Let X be a based CW-complex and P be a perfect normal subgroup of $\pi_1(X)$. Then we say a based acyclic map of CW-complexes $f : X \to Y$ is a +-construction relative to P if P is the kernel of the induced map $\pi_1(X) \to \pi_1(Y)$.*

Theorem 1 (Quillen)**.** *Let X be a based CW-complex and P be a perfect normal subgroup of $\pi_1(X)$.*

1. *There exists a +-construction $X \to Y$ relative to P.*
2. *A +-construction $f : X \to Y$ is universal among maps which kill P in the fundamental group. That is, for any $g : X \to Z$ such that the image of P in $\pi_1(Z)$ vanishes, we have a map $h : Y \to Z$, unique up to homotopy, such that $g = hf$.*
3. *In particular, if g is another +-construction relative to P, then there is a homotopy equivalence $h : Y \to \sim Z$ compatible with the maps.*

Proposition 3. *Let a map $f : X \to Y$ of based spaces be acyclic. Then it preserves integral homology wrt any local coefficient system.*

A proof of this was first given by Quillen in [], and uses topological obstruction theory. A more recent exposition is available in [].

Any group G has a maximal perfect normal subgroup. [] From now on, for any based CW-complex X, we will denote by X^+ any +-construction on X relative to the maximal perfect subgroup $P \triangleleft \pi_1(X)$.

4.1.3 Explicit Construction

A number of explicit *models* exist for the +-construction. Here is one of them which constructs X^+ by attaching 2-cells to kill the maximal perfect subgroup of $\pi_1(X)$, and then 3-cells to recover the integral homology of X.

4.2 The $BGL(R)^+$ construction for a ring R

4.3 $S^{-1}S$ construction on a Symmetric Monoidal Category

4.4 Q-construction on an Exact Category

4.5 Quillen's $+ = Q$ Theorem